
Final report for Xinje project

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Project Introduction

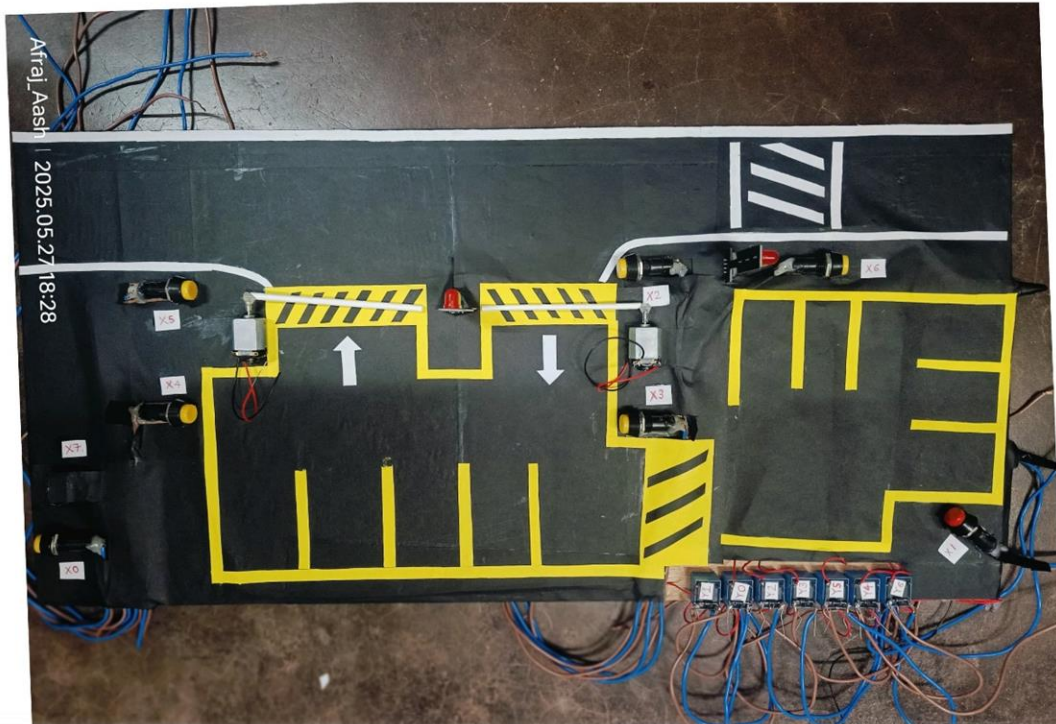
The project is based on the implementation of the parking management system using a Xinje PLC. The objective is to design and understand the working mechanism of a car park with an interface to a traffic light system.

It has a parking space for a maximum of 10 cars. A traffic light system is used to manage both vehicle access and pedestrian safety. The system works by displaying a green light when there is parking space available and a red light to signal that the car park is full.

It also includes a pedestrian crossing system. Under normal circumstances, the pedestrian traffic light is green, and vehicles can move freely. However, when the crossing button is activated by a pedestrian, the system starts the pedestrian traffic sequence to cross safely.

A ladder diagram and functional prototype were developed to simulate and demonstrate the functionality of the overall system.

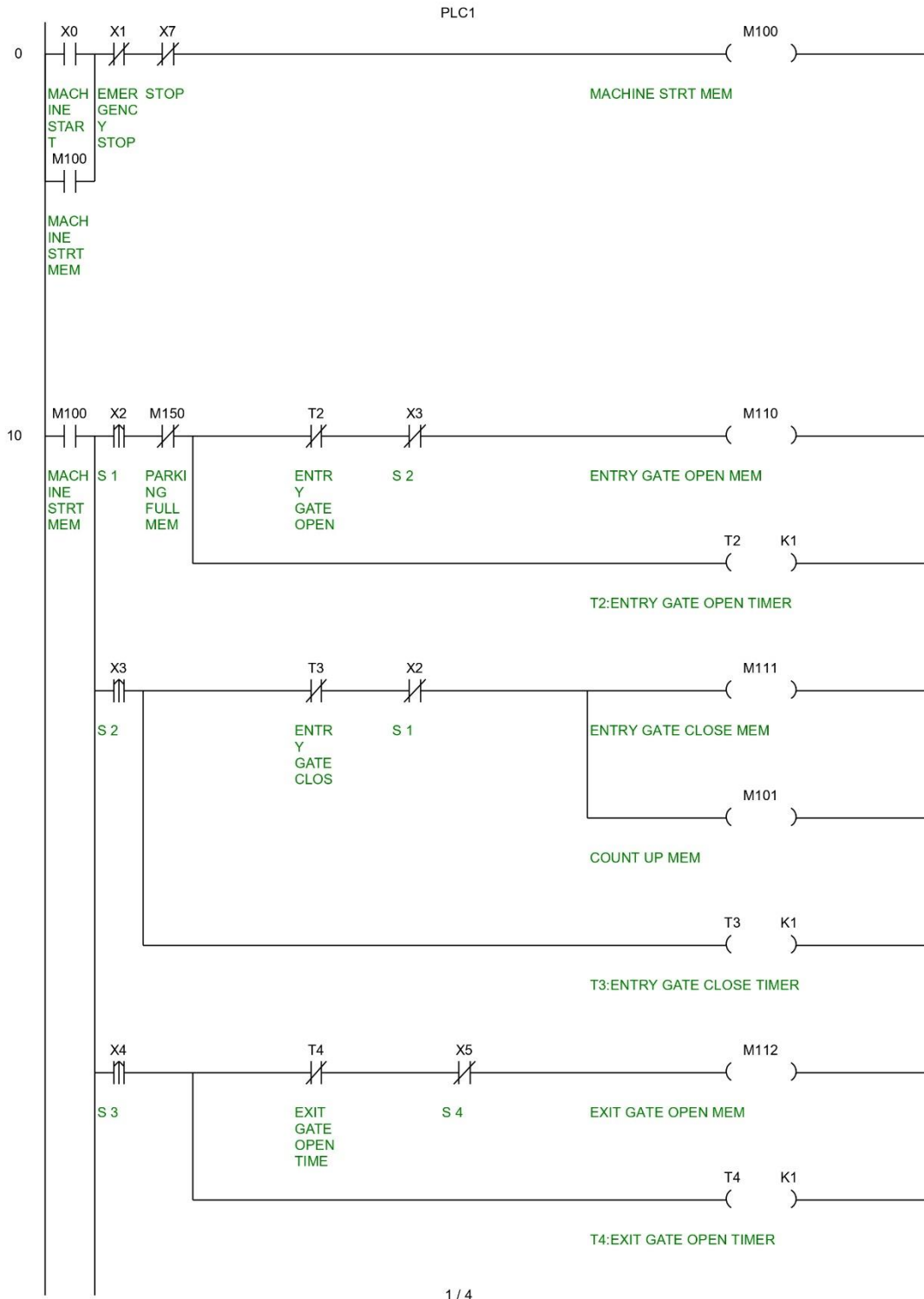
System design

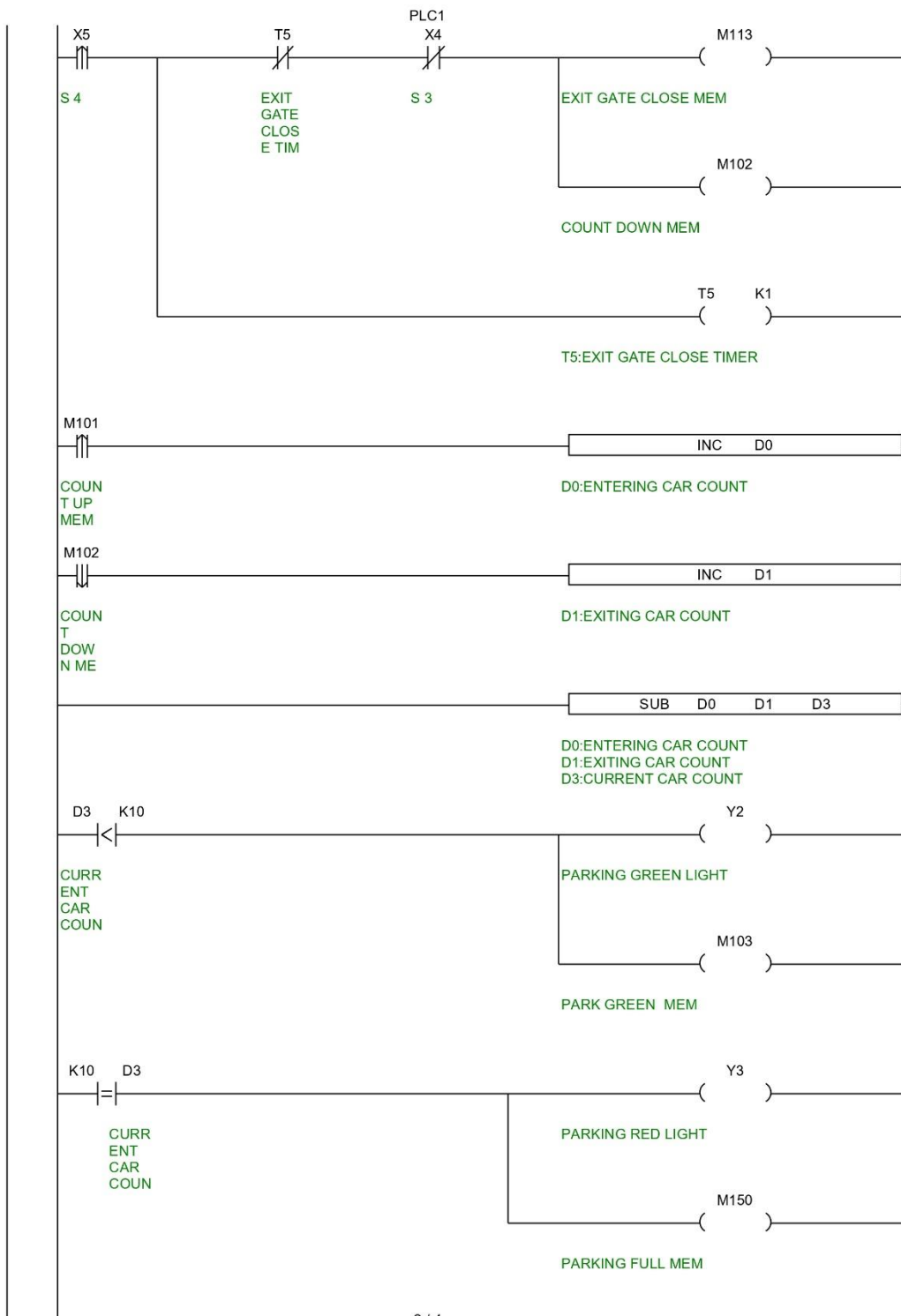


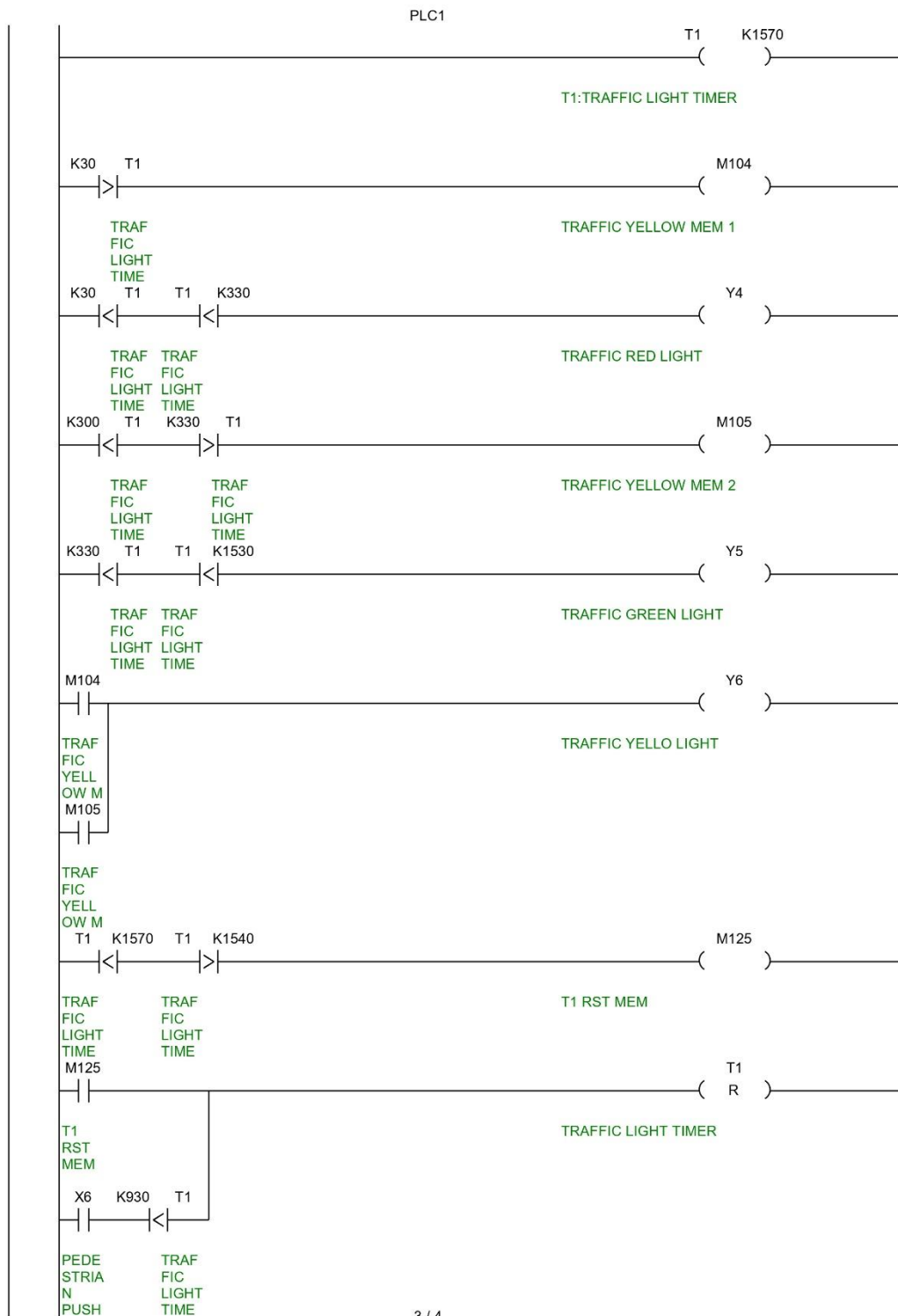
Component with justification

- Relay(24V): 5V components were used in the system. To connect that with the Xinje PLC 24V, relay used.
- Push Button for start and stop button: Memory was used to remain the machine start and normally closed switch used for emergencies.
- Sensor replaces with Push Button: 24V sensors are not cost effective for this prototype so push button was replaced with push button to identify the vehicle.
- Motor: Operate gate open and close. 5V motor was connected to the PLC with 24V relay.

Ladder logic program







I/O Mapping Table

Input	Component
X0	start push button
X1	emergency push button
X2	entry gate open sensor
X3	entry gate close sensor
X4	exit gate open sensor
X5	exit gate close sensor
X6	pedestrian crossing push button
X7	stop button

Output	Components
Y0	entry gate motor
Y1	exit gate motor
Y2	parking green light
Y3	parking red light
Y4	traffic red light
Y5	traffic green light
Y6	traffic yellow light

Logic description

This is a prototype of a car parking system with a traffic light system for pedestrian crossing. In the car parking, a total of 10 cars can be parked at a time. If the parking is full, a red light is placed at the front of the parking to indicate it, and a green light is placed to indicate slot availability for parking. The parking system has a start push button (X0), an emergency stop button (X1), and a stop button (X7).

When a car comes in front of the entry gate, the entry gate open sensor (X2) detects it and checks the space availability. If space is available, the entry gate opens and lets the car in. When the car passes the entry gate, the entry gate close sensor (X3) detects it and closes the entry gate.

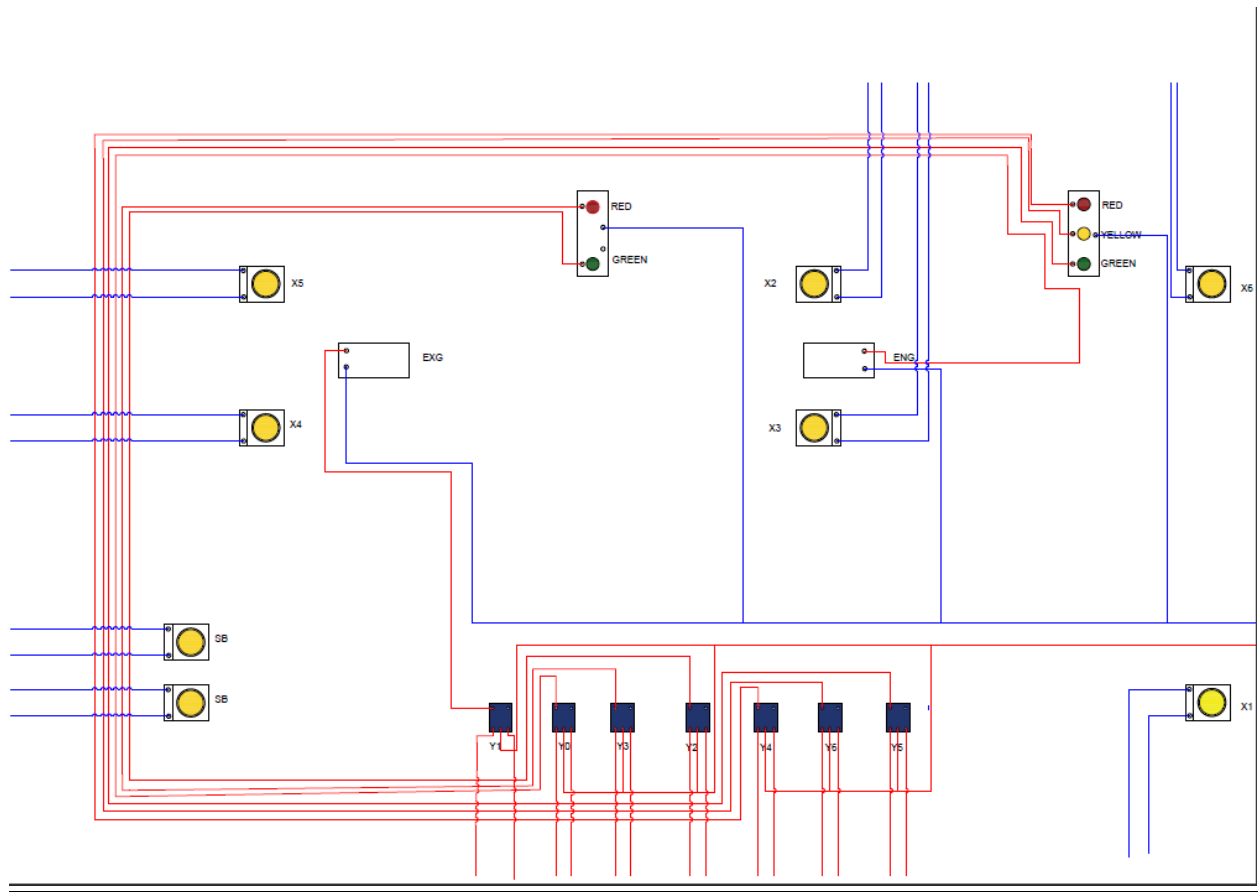
When a car goes to the exit gate, the exit gate open sensor (X4) detects the car and opens the exit gate. After passing the exit gate, the exit gate close sensor (X5) detects it and closes the exit gate.

The number of currently parked cars is calculated by the arithmetic function:
 $D3 \text{ (current car count)} = D0 \text{ (entering car count)} - D1 \text{ (exiting car count)}$.

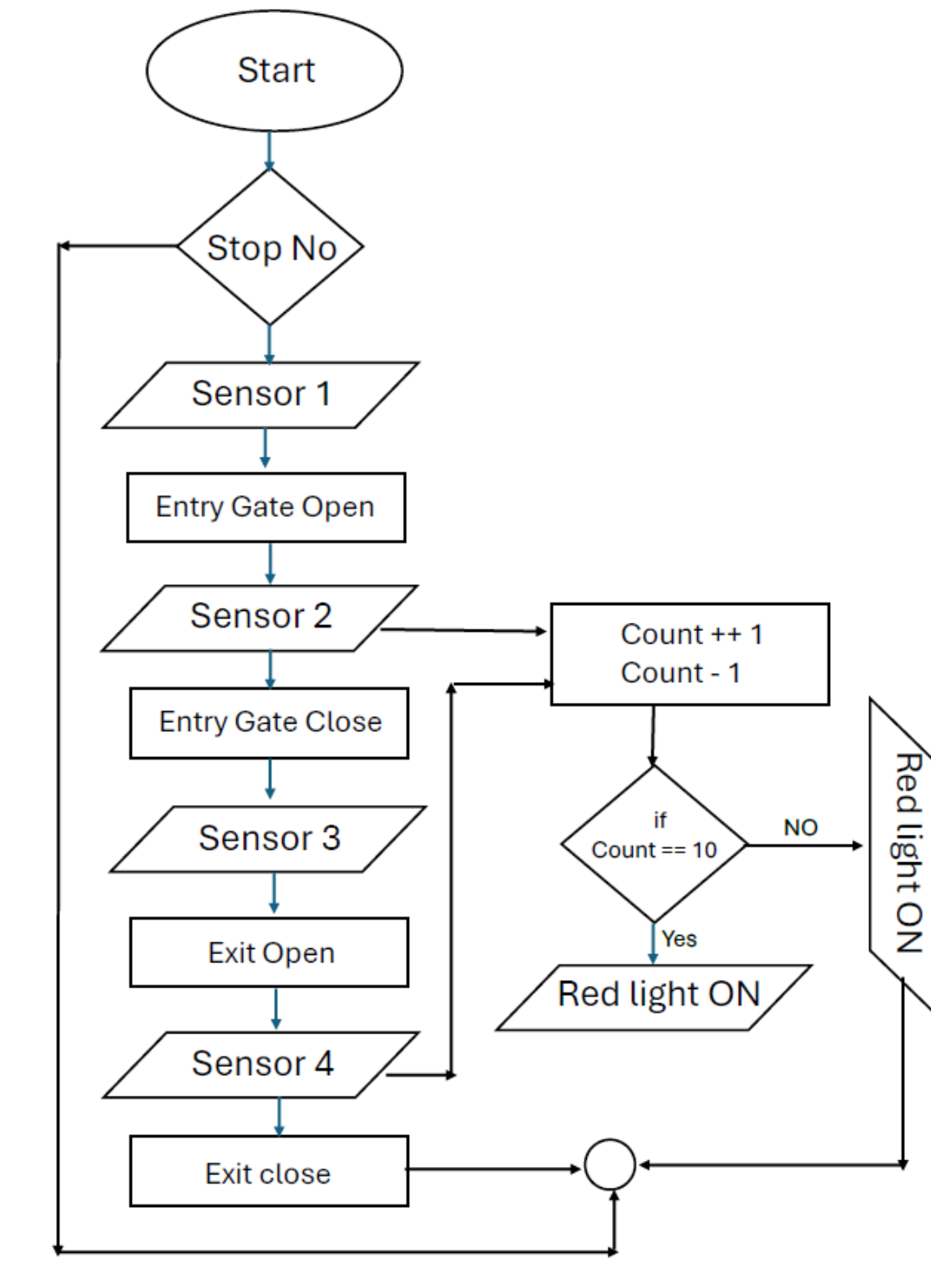
The traffic light is used for pedestrian crossing. The orange light works for 3 seconds, and the red light works for 30 seconds. After 30 seconds, the red light goes off and the green light works for 120 seconds. After this sequence, the traffic light resets to 0 seconds.

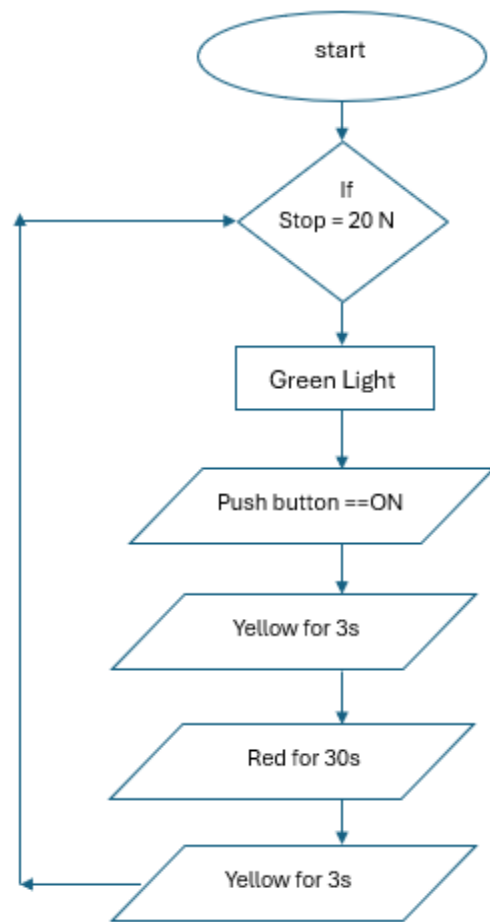
When a pedestrian comes and pushes the push button (X6), the algorithm checks whether the push button was pressed after 93 seconds from the starting time of the traffic light. If it is true, the green light goes off and the red light turns on, allowing the pedestrian to cross the road.

Wiring diagram

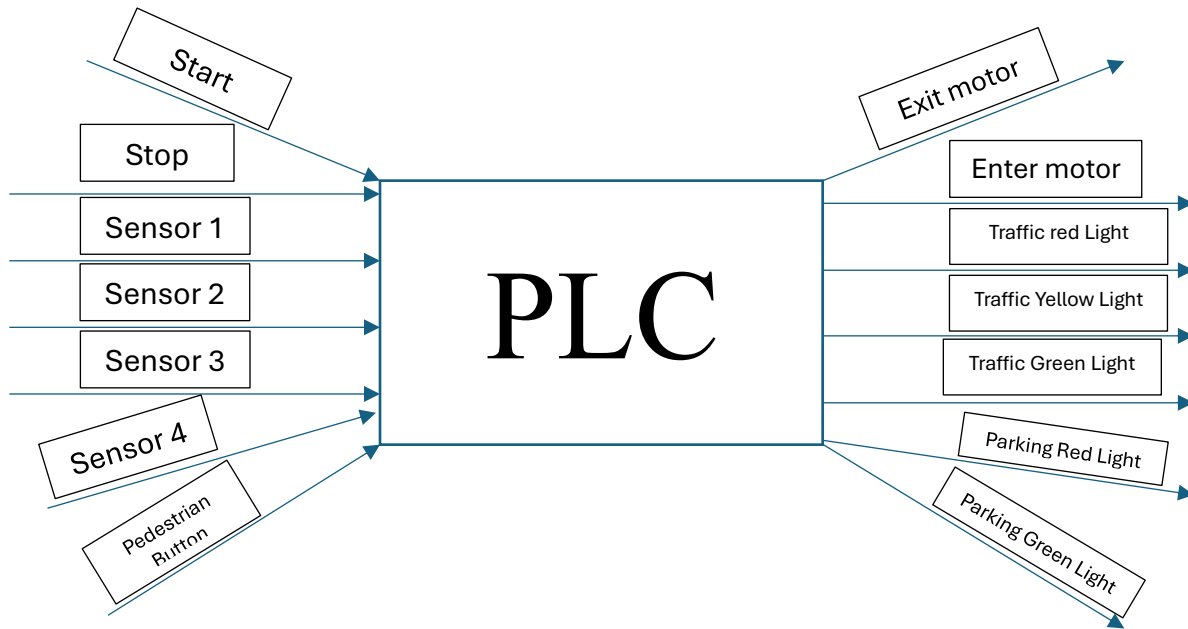


Flow chart of the system





Schematic diagram



Testing results



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Assumption

- Push button used for the replacement of sensors.

Results and Discussion

The car parking system prototype with traffic light for pedestrians was successfully designed, programmed, and tested on a Xinje PLC. The system was able to demonstrate the automatic control of vehicle entry and exit and the deployment of a traffic light system for pedestrian safety.

The following key results were achieved during system testing:

Parking Slot Management:

- The system was counting the number of cars coming in and going out of the parking lot correctly.
- Arithmetic logic designed with the use of data registers (D0, D1, and D3) was correctly keeping the updated count of vehicles.
- Red and green LEDs at the parking entrance responded accordingly based on slot availability. When the parking had 10 cars filled, the red light illuminated, preventing any further entrance.

Gate Control Functionality:

- The (X2, X3) entrance and (X4, X5) exit sensors worked as expected to close and open the gates based on vehicle presence.
- There were no delays or failures in gate operation, and the gates offered flawless and automatic control with no need for manual override.

Pedestrian Traffic Light Operation:

- Traffic light operation was on the expected time cycle: 3 seconds for orange, 30 seconds for red, and 120 seconds for green.
- When a pedestrian pressed the push button (X6) after 93 seconds of green light activation, the system was successfully able to cut off the green phase and switch on the red light. This allowed safe pedestrian crossing.
- The system then resumed its normal traffic light cycle, which meant the reset and timing circuits were working as needed.

Safety and Emergency Controls:

- Emergency stop button (X1) and stop button (X7) were tested and confirmed to prioritize the system operation for safety in unexpected situations.

Overall, the prototype functioned as expected and met the requirements of the design. It demonstrated how one can go about using programmable logic controllers (PLCs) for the automation of real-life systems that have a number of inputs and outputs, such as parking control and management of pedestrian traffic. Minimal delays in sensor response and light changeovers were noticed but did not impact the overall performance of the system.

Group member contributions

Prototype design- All 6

Prototype creating - Premnath, Sathu, Sujaith and Basura

Ladder program design- Afraj Aash and Premnath

Get things for project- Napithan, Sathu, Premnath, Sujaith and Afraj Aash

Create PowerPoint and report- Afraj Aash and Basura

Wiring diagram CAD- Basura and Sujaith

Flow chart- Napithan and Sujaith

System design -Premnath, Napithan, Basura and Sujaith